EcoLoop Building Agents

AI Closed-Loop Building Energy Control

Problem Statement ID	[INSERT FROM PORTAL]
Problem Statement Title	Eco-Loop Building Agents
Theme	Physical AI / Autonomous Smart Building Energy Optimization
PS Category	Software
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Student ID	[INSERT FROM PORTAL]

Proposed Solution

1

A Real Closed Loop

An open-source LLM reads live EnergyPlus sensor data and injects new HVAC setpoints back into the running simulation — not a scripted demo.

2

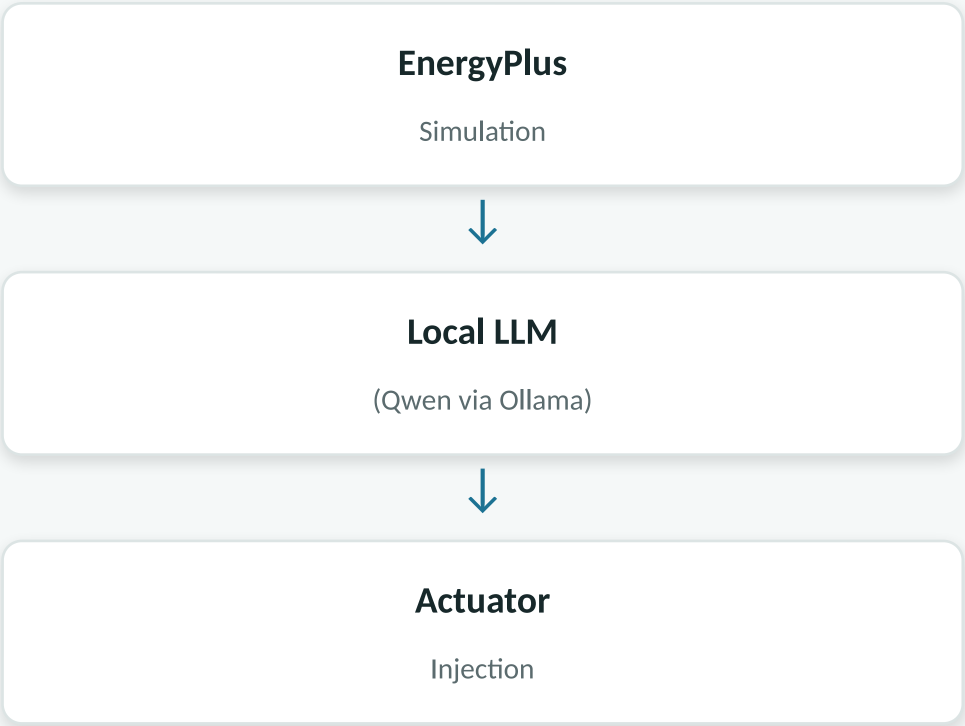
Adaptive, Not Static

EnergyPlus streams zone temperatures and cumulative energy use to a local LLM every control cycle, which reasons about comfort vs. efficiency and returns new setpoints written back live via the actuator API.

3

Safety-Constrained by Design

Comfort bounds are hard-clamped in code, not just prompted — the LLM cannot push occupants outside a safe range even if it hallucinates a bad value.



closed loop — repeats every control cycle

Technical Approach

EnergyPlus Simulation → **Sensor Data** API → **LLM Reasoning** (Qwen / Ollama) → **Tool-Calling** Engine → **Actuator** Injection

Technology Stack

- Python — orchestration, tool-calling loop
- EnergyPlus 26.1 — physics simulation via the pyenergyplus Data Transfer API
- Ollama + Qwen2.5-7B — local LLM, no cloud dependency
- eppy — programmatic IDF authoring
- Streamlit — live savings dashboard

Control Cycle

- Feedback: zone temps + facility kWh read every timestep
- Reasoning: LLM consulted every Nth step to bound latency
- Control: clamped setpoints written back via set_actuator_value
- Custom JSON tool-calling loop in place of a full MCP server — lower integration risk under time pressure

Feasibility & Viability

1

Feasibility

Every layer runs on free, open tooling — EnergyPlus (public domain, DOE), Ollama + Qwen2.5 (open-weight, local), Python. Proven end-to-end on a single laptop with no cloud dependency.

2

Challenges & Risks

Local LLM latency/timeouts mid-simulation; small models occasionally return malformed JSON; a naive setpoint override could sacrifice comfort for energy savings.

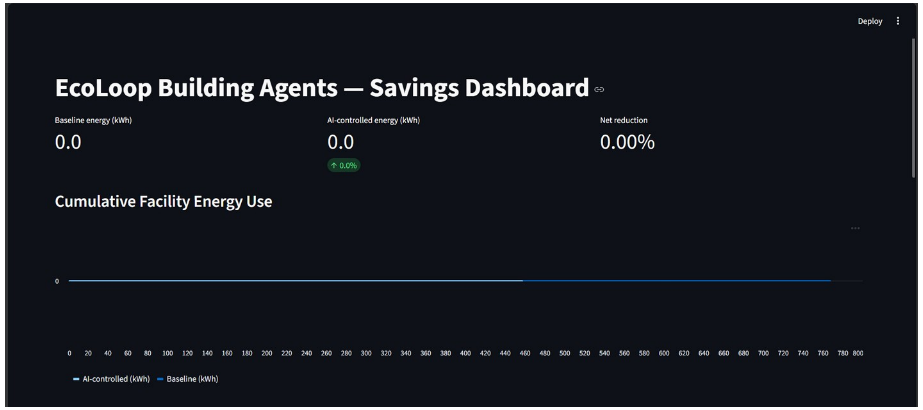
3

Mitigation Strategy

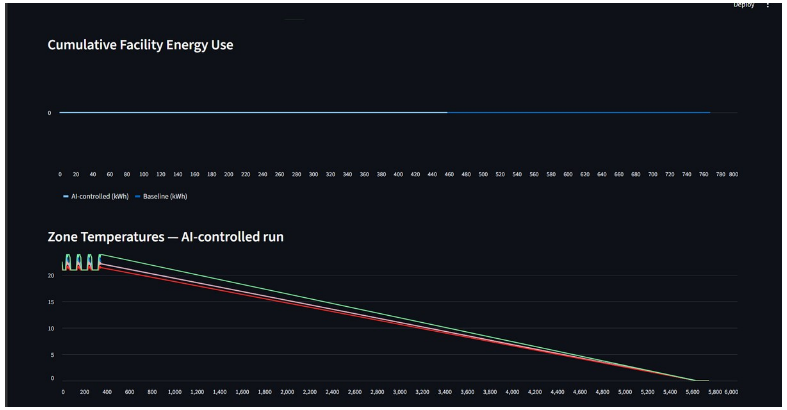
Fallback-and-hold logic keeps the previous setpoint on a failed/unparseable call rather than crashing the sim. Comfort bounds are hard-clamped in code, not just prompted.

Artifacts — Live System Proof

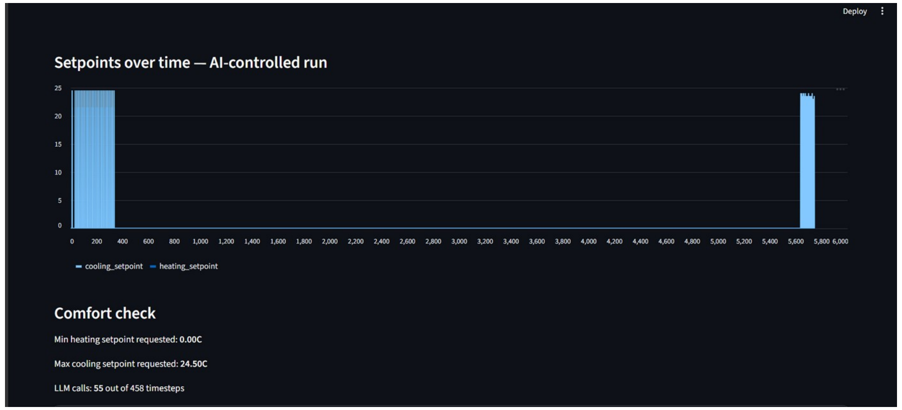
Repository: https://github.com/shenal19/Eco-Loop-Building-Agents-HoneyWell-Assessment-



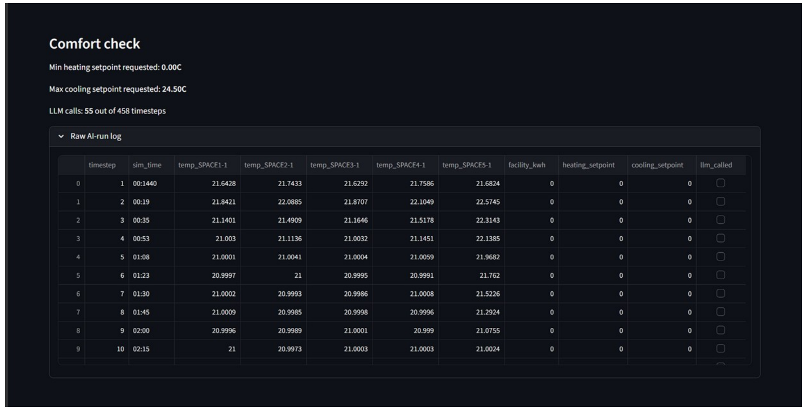
Savings Dashboard — KPIs & Cumulative Energy



Energy Use & Zone Temperatures (AI-controlled run)



Setpoints Over Time & Comfort Check



	timestep	sim_time	temp_SPACE1-1	temp_SPACE2-1	temp_SPACE3-1	temp_SPACE4-1	temp_SPACE5-1	facility_kwh	heating_setpoint	cooling_setpoint	llm_called
0	1	00:1440	21.6428	21.7433	21.6292	21.7586	21.6824	0	0	0	☐
1	2	00:19	21.8421	22.0885	21.8707	22.1049	22.5745	0	0	0	☐
2	3	00:35	21.1401	21.4909	21.1646	21.5178	22.3143	0	0	0	☐
3	4	00:53	21.003	21.1136	21.0032	21.1451	22.1385	0	0	0	☐
4	5	01:08	21.0001	21.0041	21.0004	21.0059	21.9682	0	0	0	☐
5	6	01:23	20.9997	21	20.9995	20.9991	21.762	0	0	0	☐
6	7	01:30	21.0002	20.9993	20.9986	21.0008	21.5226	0	0	0	☐
7	8	01:45	21.0009	20.9985	20.9998	20.9996	21.2924	0	0	0	☐
8	9	02:00	20.9996	20.9989	21.0001	20.999	21.0755	0	0	0	☐
9	10	02:15	21	20.9973	21.0003	21.0003	21.0024	0	0	0	☐

Raw AI-Run Log (per-timestep control trace)

Research & References

- EnergyPlus documentation & Python API (Data Transfer / EMS actuators) — energyplus.net
- eppy library documentation (IDF scripting) — github.com/santoshphilip/eppy
- Ollama model & API documentation (local LLM serving) — ollama.com

Thank You

Shenbagabalaji A • EcoLoop Building Agents • Physical AI Building Sandbox PoC

Questions welcome.